Python Activity-Based Learning
Case Study

12 Attendance Recovery Computation

Selected case study | Prepared using the topics listed in the supplied

assignment. ## 1. Problem Statement

Calculate a student's current attendance percentage and determine the minimum number of consecutive classes the student must attend to reach a required attendance percentage.

2. Objectives

• Calculate attendance percentage.

• Use a loop to find the minimum classes required.

• Apply conditions to classify attendance eligibility.

• Display a clear result.

3. Assumptions

• Conducted classes and attended classes are non-negative integers.

• Attended classes cannot exceed conducted classes.

• The eligibility target is 75%.

• The student attends every future class.

4. Algorithm

1. Input conducted classes and attended classes.

2. Validate the inputs.

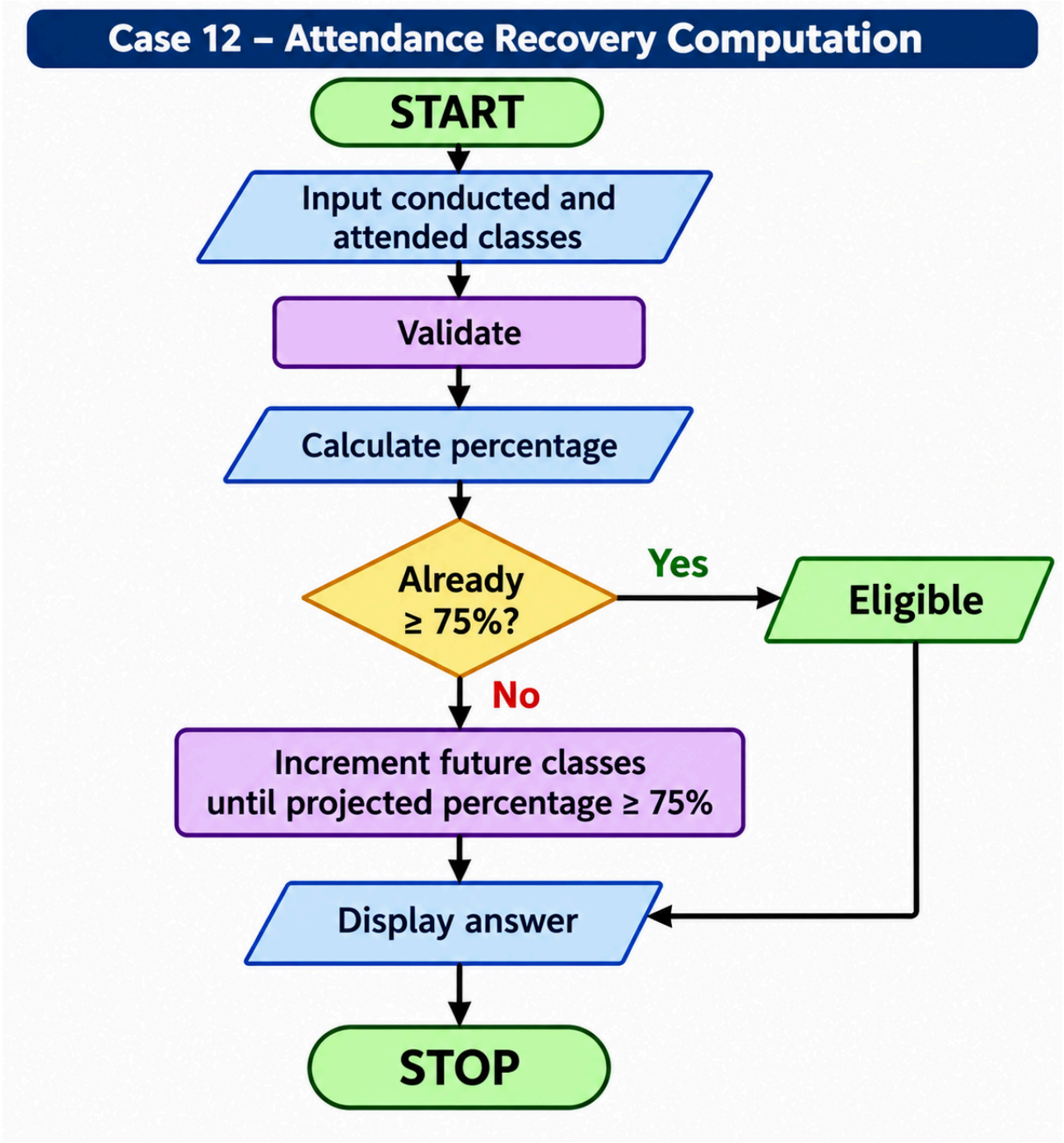
3. Calculate current attendance percentage.

4. If the percentage is already at least 75%, report eligible.

5. Otherwise, start future class count at zero and repeat until the projected percentage reaches

75%. 6. Display the minimum future classes required.

5. Flowchart



6. Pseudocode

```
BEGIN
 INPUT conducted, attended
 IF inputs invalid THEN PRINT error
 ELSE
 percentage ← attended / conducted × 100
 future ← 0
 WHILE (attended + future) / (conducted + future) × 100 < 75
 future ← future + 1
 END WHILE
```

```
 PRINT percentage, future
 END IF
END
```

Case Study 12 — Attendance Recovery Computation

7. Python Source Code

```
conducted = int(input("Total classes conducted: "))
attended = int(input("Classes attended: "))

if conducted <= 0 or attended < 0 or attended > conducted:
 print("Invalid attendance details")
else:
 percentage = attended / conducted * 100
 print("Current attendance:", round(percentage, 2), "%")

 future = 0
 while ((attended + future) / (conducted + future)) * 100 < 75:
 future += 1

 if future == 0:
 print("You are already eligible.")
 else:
 print("Minimum additional classes to attend:", future)
 print("Projected attendance:",
 round((attended + future) / (conducted + future) * 100, 2), "%")
```

8. Test Cases

Input / Scenario	Expected result
Conducted 100; attended 60	Current 60%; attend next 60 classes to reach 75%
Conducted 40; attended 32	Current 80%; already eligible; 0 classes
Conducted 50; attended 35	Current 70%; attend 10 classes to reach 75%

9. Conclusion

The program uses arithmetic and a while loop to calculate attendance recovery requirements. Note: Run

the program in Python 3. The sample test cases are illustrative; enter the values when prompted.